Corso di Laurea in Ingegneria Informatica Ingegneria del Software e Progettazione Web A.A. 2025/2026

Technical Documentation

NightAgent

Events and Shows Management Software

Professors

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Students

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1 Software Requirements Specification

1.1 Introduction

1.1.1 Aim of the document

The aim of this document is to outline the functional and non-functional specifications for the NightAgent software. It provides a comprehensive vision of the software's features and highlights any discrepancies between the final project and the initial prototype. This document serves as a crucial reference for developers, testers, and project stakeholders, ensuring a clear understanding of the project's scope and requirements.

Additionally, it details the fundamental hardware and software requirements necessary for NightAgent's operation. Included in this document is a link to our SonarCloud dashboard, which grants access to our GitHub repository. From this dashboard, you can review the analysis of bugs, code smells, and security vulnerabilities within the software, as well as the steps taken to address these issues throughout the development process.

The source code, along with all other deliverables, is available in the attached .zip folder. This comprehensive package provides everything needed for a total understanding and evaluation of the NightAgent project.

1.1.2 Overview of the defined system

NightAgent is a standalone desktop application designed to assist users in discovering local events and making new friends. It provides a unified platform where users can browse all organized events in their city. Additionally, NightAgent offers event organizers a powerful tool to reach a broad audience without the need to promote their events through other channels such as social media, websites, or television.

This centralized approach makes it easier for both participants and organizers, improving the overall experience and community engagement.

1.1.3 Hardware and software requirements

Hardware Requirements

To ensure optimal performance of the NightAgent application, the following hardware specifications are recommended:

Processor: Intel Core i3 or equivalent AMD

Memory (RAM): 2 GB minimum

Display: 1280 x 800 resolution or higher

Software Requirements

The NightAgent application requires the following software components:

Operating System: Windows 10 or higher, macOS 10.14 or higher, Linux (kernel 3.10 or higher)

Java Runtime Environment (JRE): JRE 17 or higher

These hardware and software requirements ensure that NightAgent runs efficiently and provides a good user experience.

1.1.4 Related systems, pros and cons

Meetup: This platform enables users to organize and participate in events based on their interests. Users can join specific groups and attend events organized by these groups. However, Meetup restricts participation to group-specific events. In contrast, NightAgent allows users to access and participate in all events organized in their city, providing a more comprehensive event discovery experience.

Eventbrite: This event management platform allows organizers to create and promote events, while users can discover and participate in both local and online events. However, Eventbrite does not support the creation of groups for social interaction. NightAgent, on the other hand, allows users to create and join groups where they can chat and get to know each other, creating a sense of community and making event participation more engaging.

1.2 User Stories

1.2.1 Matteo Della Giovampaola:

1. As a user, I want to be able to see all the events in my city, so that I don't have to search from multiple sources*.

2. As a user, I want to be able to book my attendance for an event, so that I am sure I can attend the event.

3. As an organizer, I want people to know when I post a new event in their city, so that I can have more customers.

*sources: social media pages, webpages

1.2.2 Emanuele Moscetta:

1. As a user, I want to be able to chat with other users participating to the same event, to make new friends.

2. As an organizer, I want to be able to see analytics* about my past events, so that I can improve the quality and organization of future events.

3. As an organizer, I want to be able to edit an event, so that I can keep users informed about changes.

*analytics: times event was clicked by users, planned participants, actual participants

1.3 Functional requirements

1.3.1 Matteo Della Giovampaola:

1. The system shall display available events in user's city.

2. The system shall show photos and information provided by the organizer in the place's page. 3. The system shall provide a past events archive for the organizer.

1.3.2 Emanuele Moscetta:

1. The system shall provide a group chat with users participating to the same event.

2. The system shall provide an interface to add, update or delete events created by the orga- nizer.

3. The system shall provide an analytics page for the event organizer.

1.4 Use cases

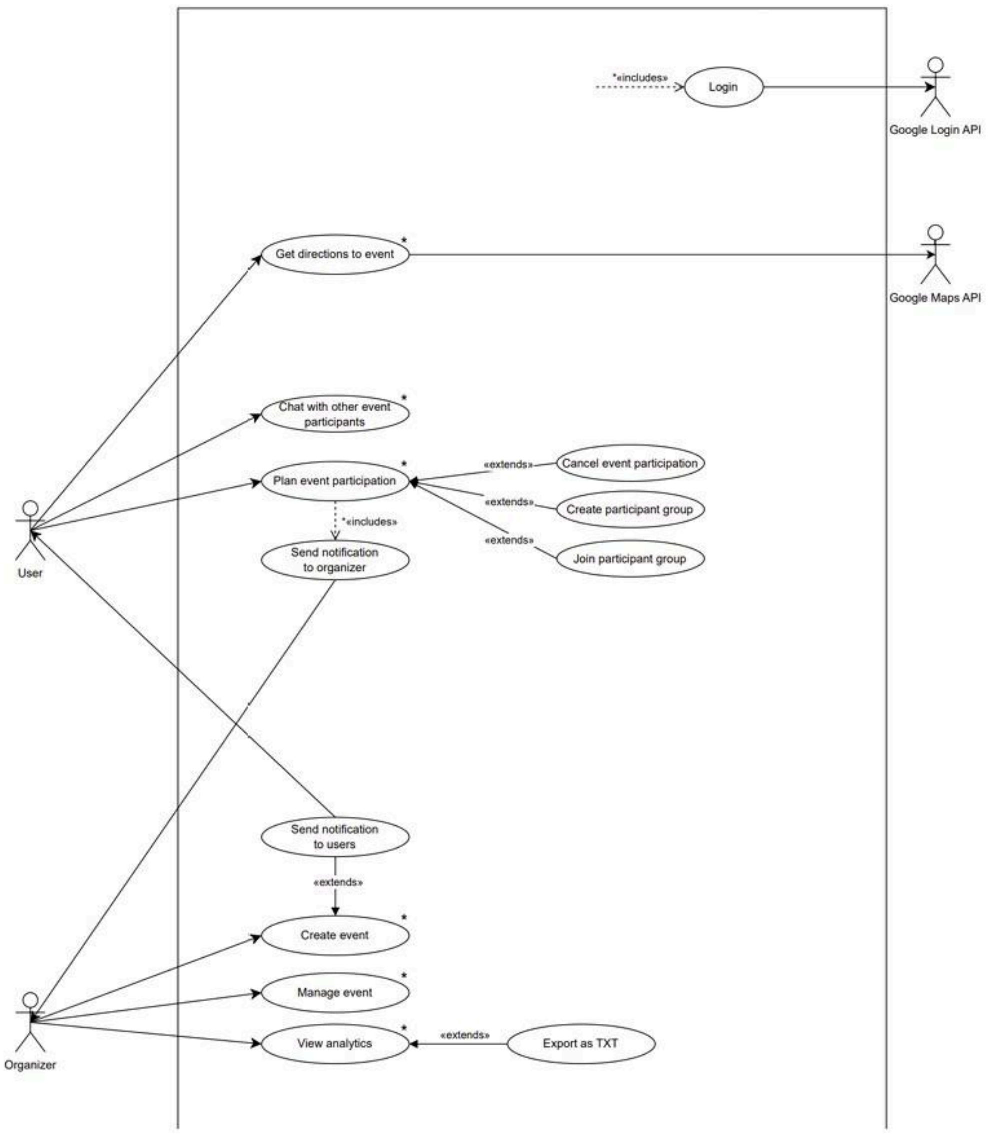
Division of Work on Deliverables

In our project, the two group members divided the work based on different use cases. Nicolas Oberi and Matteo Trossi each took responsibility for specific aspects of the project:

Matteo Della Giovampaola: has worked on the Plan Event Participation use case.

Emanuele Moscetta: has worked on the Create Event use case.

1 Overview diagram



2 Internal steps

Matteo Della Giovampaola:

Use case: *Plan event participation*

1. The system retrieves all the available events in the user's city from the system catalogue. 2. The user selects an event.

3. The system shows the user the information about the event.

4. The system asks the user if he wants to participate.*

5. The user accepts to participate.

6. The system notifies the user his participation is confirmed.

7. The system asks the user if he wants to create (or join) an event group.* 8. The system notifies the event organizer of a new participation to his event. *Extensions*:

1a: System catalogue doesn't respond: notify the user and end the use case. 1b: No events available in user's city: notify the user and end the use case. 5a: The user doesn't accept to participate: go back to the home page and end the use case. * := discrepancies

Emanuele Moscetta:

Use case: *Create event*

1. The system verifies if the user is logged as an organizer.

2. The organizer selects Add a new event .

3. The system prepares a blank creation form.

4. The organizer adds all the information in the blank fields.

5. The system uses Google Maps API to verify the correctness of the location address. * 6. The organizer completes the creation of the event.

7. The system connects to the internal database and updates it with new event data. 8. The system shows a confirmation popup organizer of the successful creation of the event. 9. The system notifies all users in event city that a new event is available. *Extensions*:

4a. Input text is too long: The system asks the organizer to insert a shorter value in the specific text field.

4b. Invalid time format: The system asks the organizer to insert a valid time. 4c. Selected date is before the current date or empty: The system asks the organizer for a valid date.

5a. Google Maps API nonresponding: The system notifies the organizer that the API is not responding and goes back to step 3. *

6a. Event with the same name already created: System notifies the organizer to change name.

7a. Database failure: The system notifies the organizer that the creation failed and terminates the use case.

Dictionary:

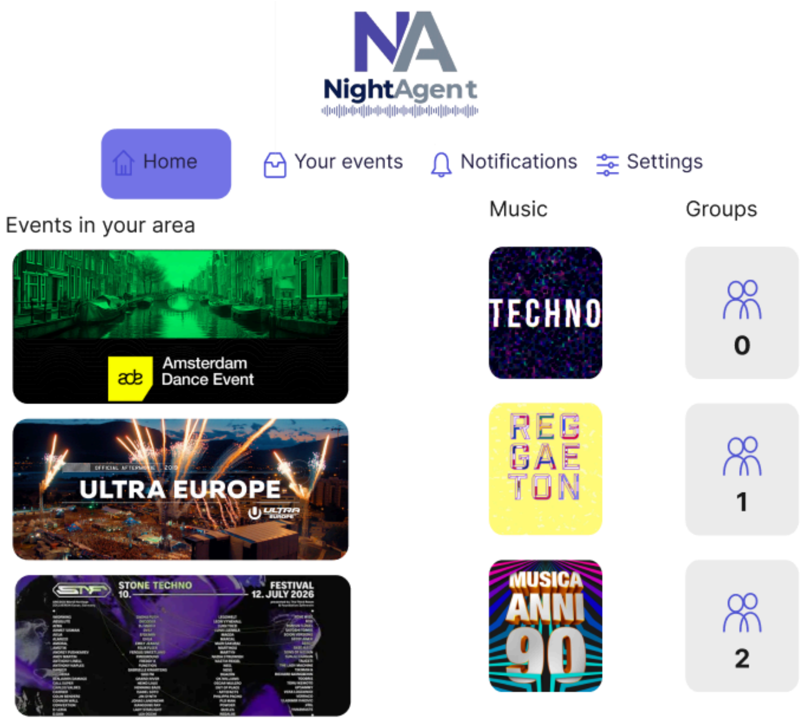
1. Information: event name, address, province, city, date from the calendar, time, type of music from a preloaded list, an image picked from user files.

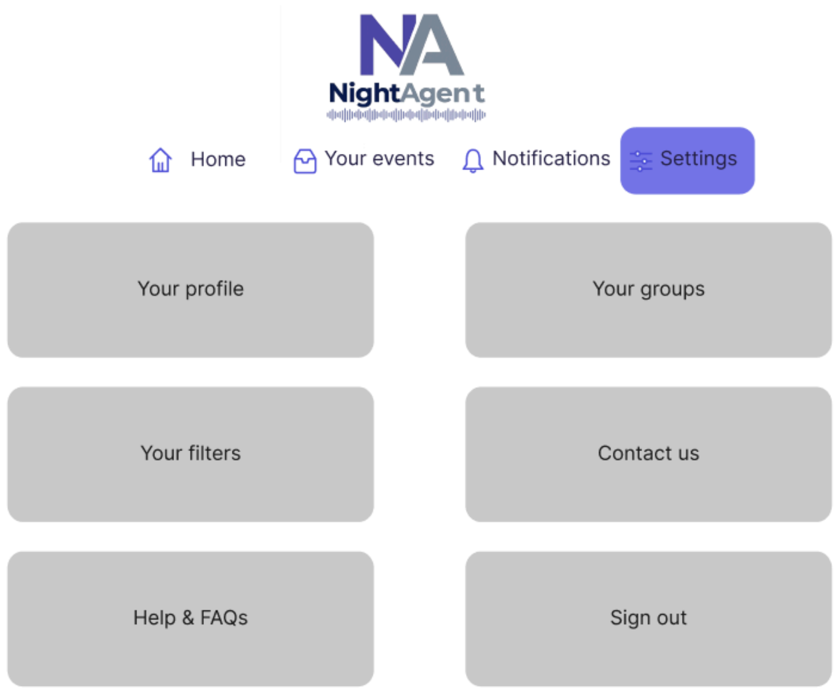
* := discrepancies

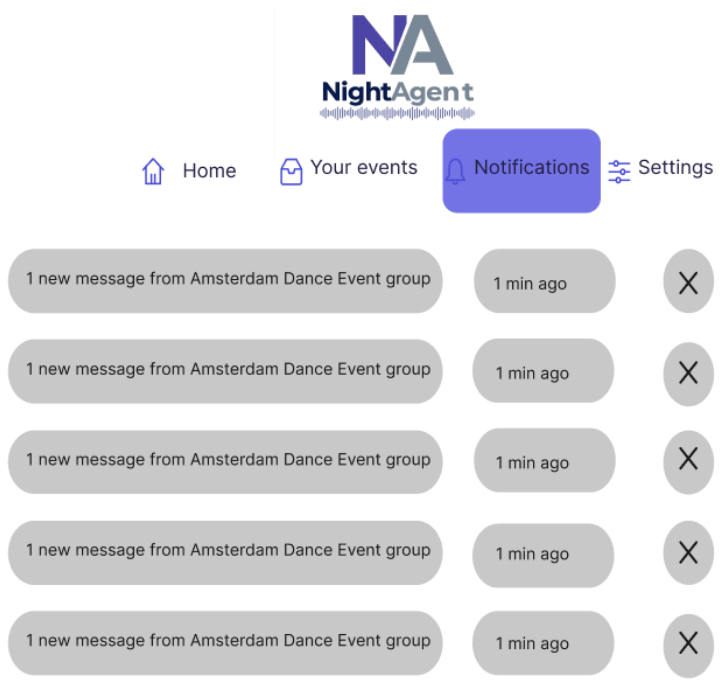
2 Storyboards

We designed three storyboards for each member of the work group using Figma online design tool. 5

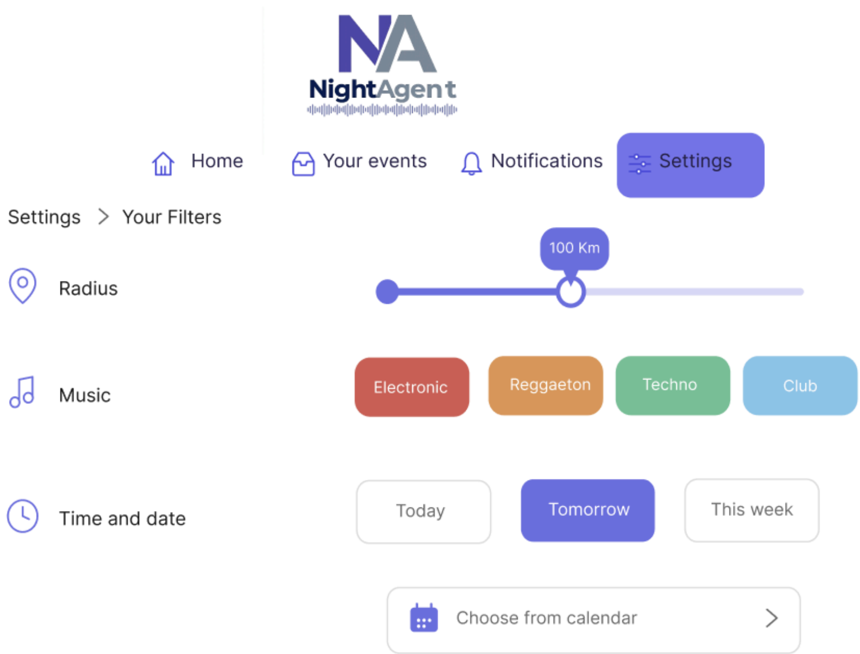
Matteo Della Giovampaola:

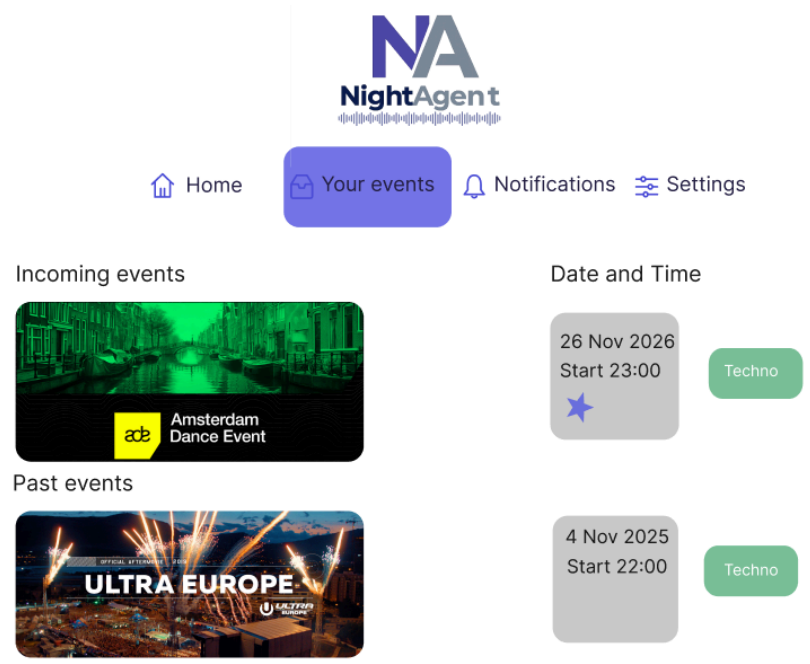


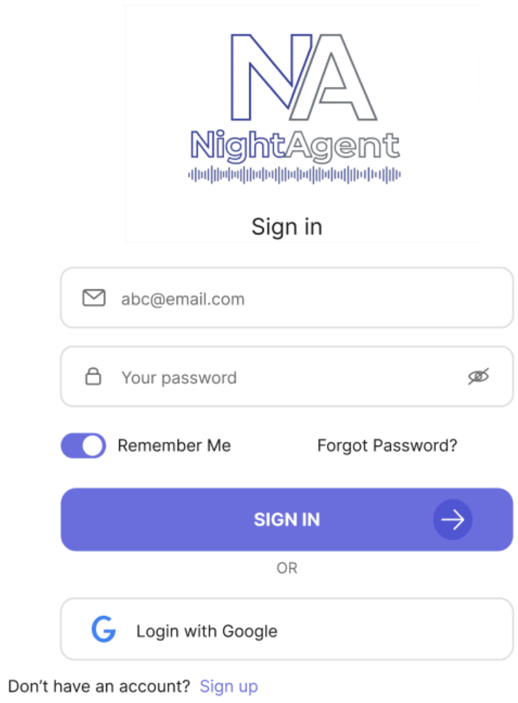




Emanuele Moscetta:




NightAgent
Home
Your events
Notifications
Settings
Incoming events
Date and Time
Amsterdam
Dance Event
26 Nov 2026
Start 23:00
Techno
Past events
ULTRA EUROPE
4 Nov 2025
Start 22:00
Techno


NightAgent
Sign in
abc@email.com
Your password
Remember Me
Forgot Password?
SIGN IN
OR
Login with Google
Don't have an account? Sign up

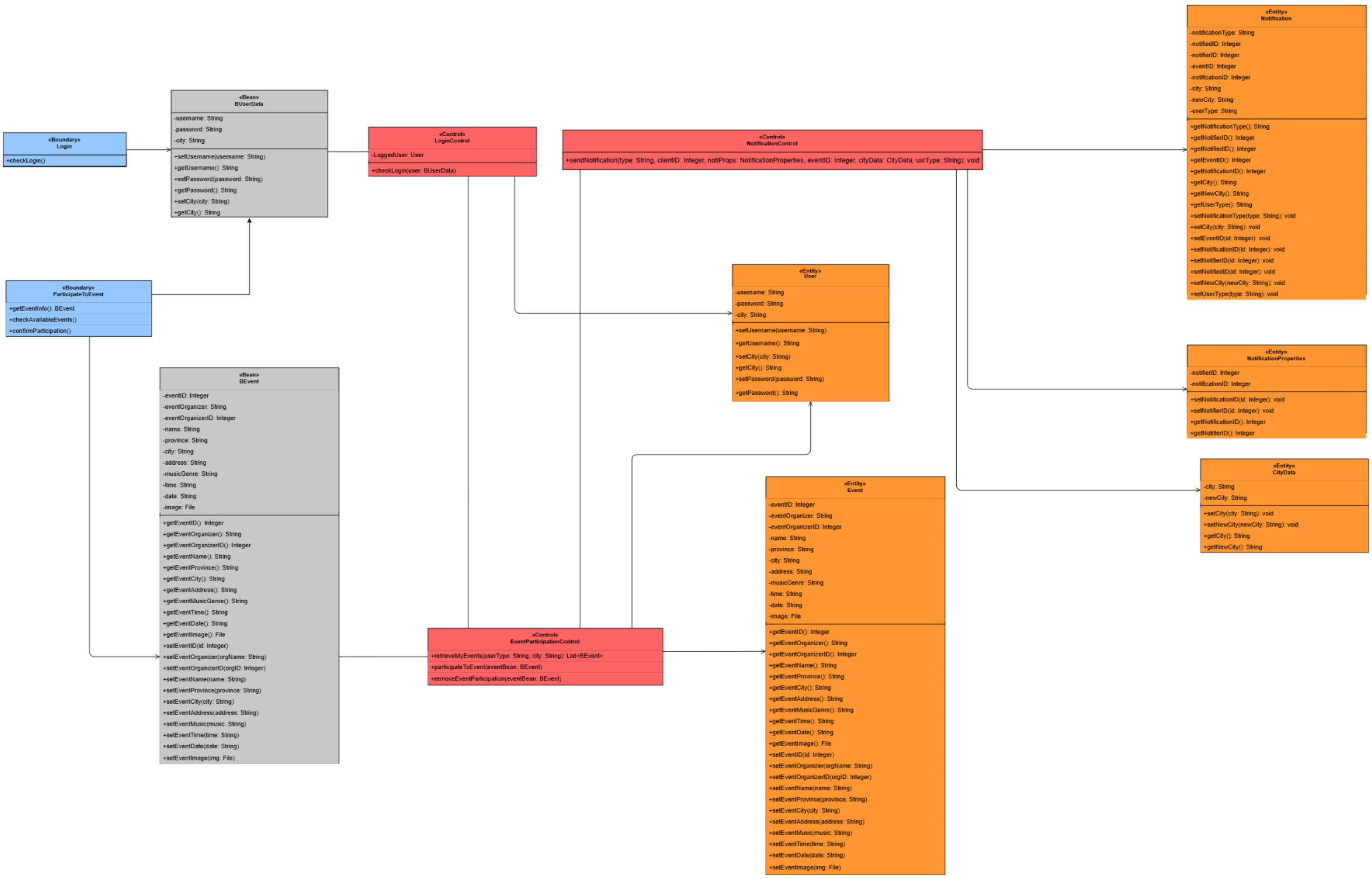
3 Design

3.1 Class Diagram

3.1.1 VOPC (View of participating classes)

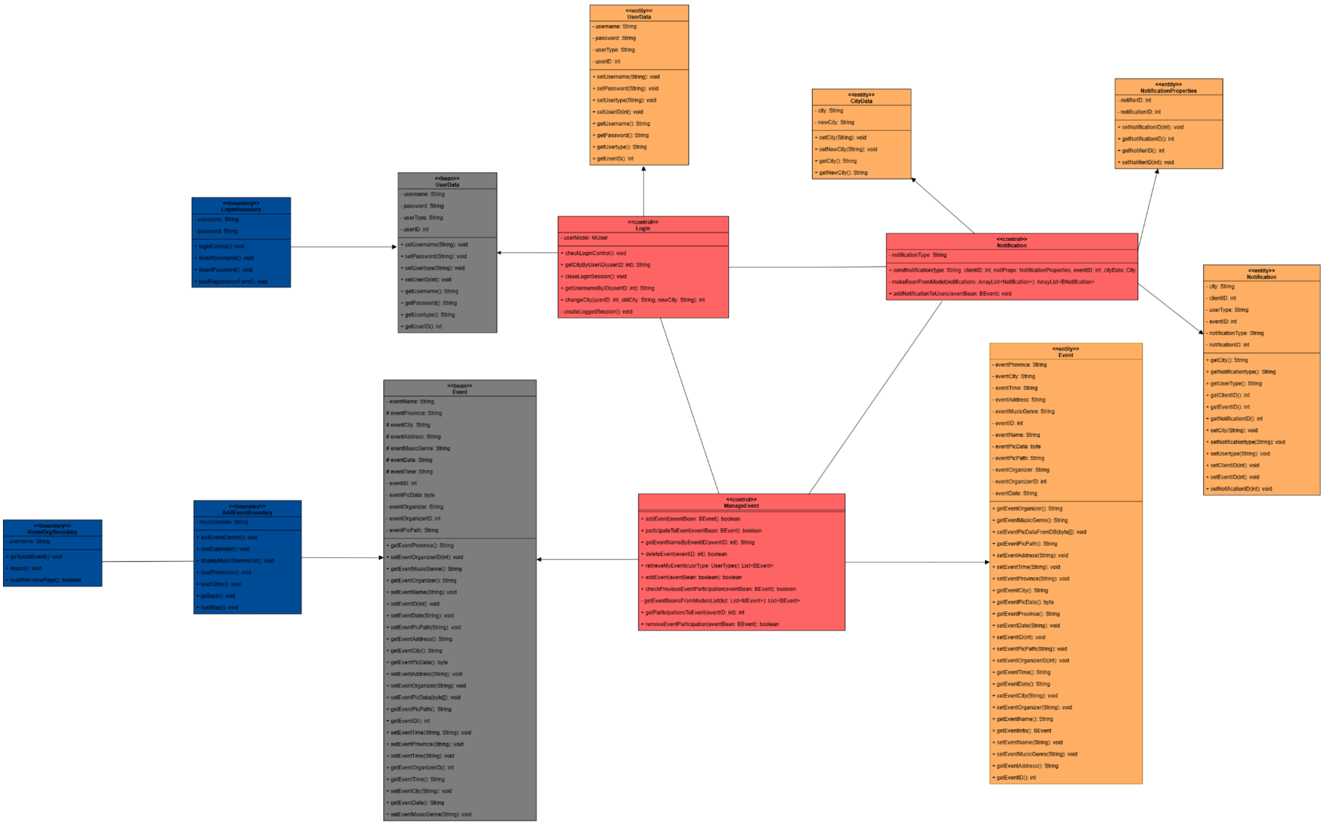
Matteo Della Giovampaola:

Use Case: *Plan event participation*



Emanuele Moscetta:

Use Case: *Create event*



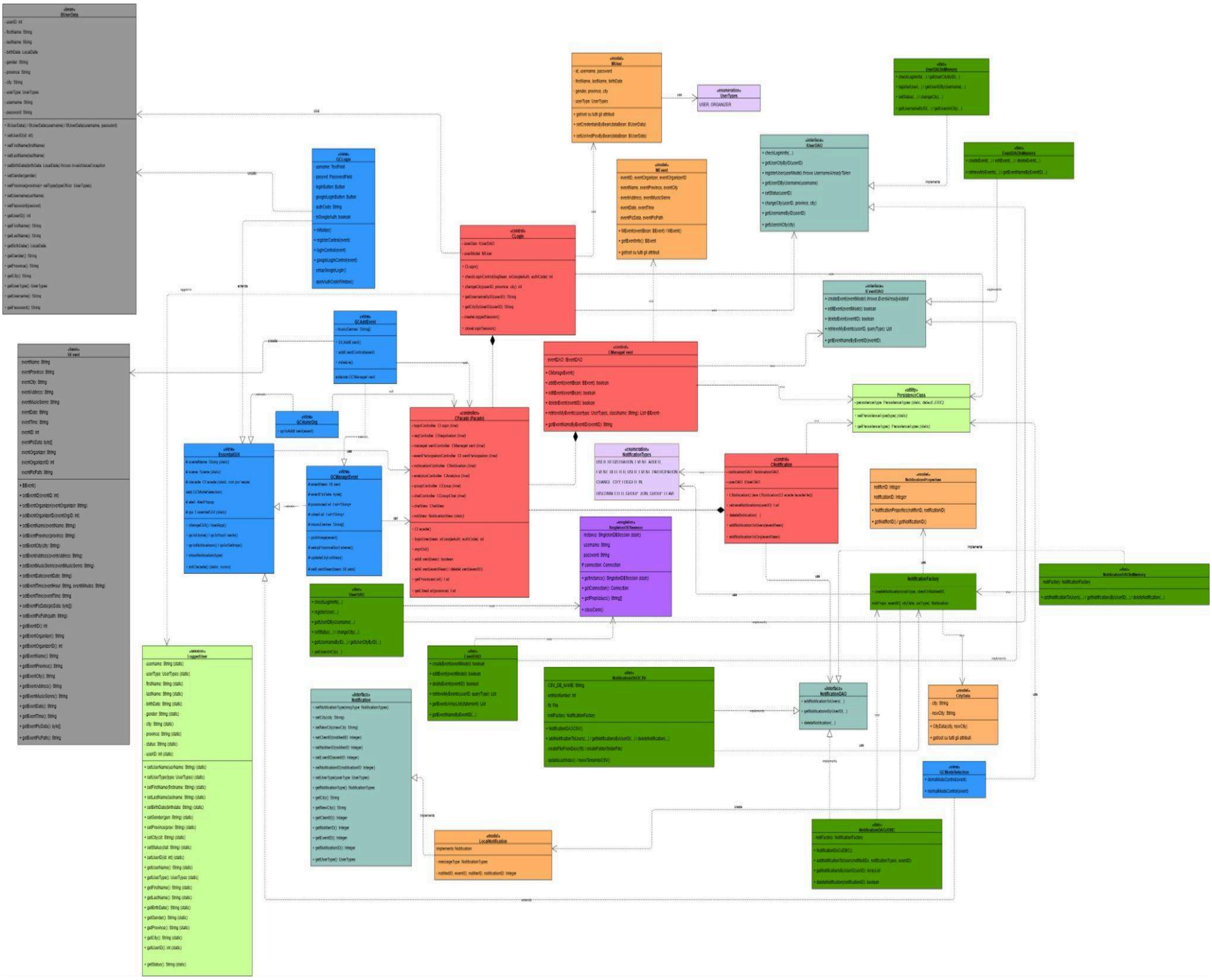
3.1.2 Design-Level Diagram

Matteo Della Giovampaola:

Use Case: *Plan event participation*

Emanuele Moscetta:

Use Case: *Create event*



3.2 Design Patterns

Within the NightAgent application, we employed various design patterns to enhance code structure and simplify specific tasks.

Singleton Pattern for Database Connections

We implemented the Singleton pattern to ensure the existence of a single instance of the database connection. This allowed us to avoid inconsistencies and duplications in data access operations performed by DAO classes. With this pattern, all classes requiring database access can do so through a single instance of connection, without needing to pass the connection as a parameter in every constructor.

Factory Pattern for Notifications

Emanuele Moscetta led the implementation of the Factory pattern, which was utilized for creating chat messages and notifications within the NightAgent application.

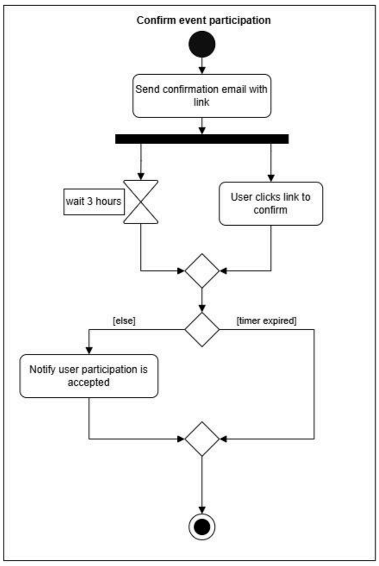
Facade Pattern for Unified Controller Management

Additionally, we applied the Facade pattern for overall management of controller methods exchange between graphical and application controllers. This pattern simplified the interface between different components of the application, enhancing code readability and maintainability.

3.3 Activity Diagram

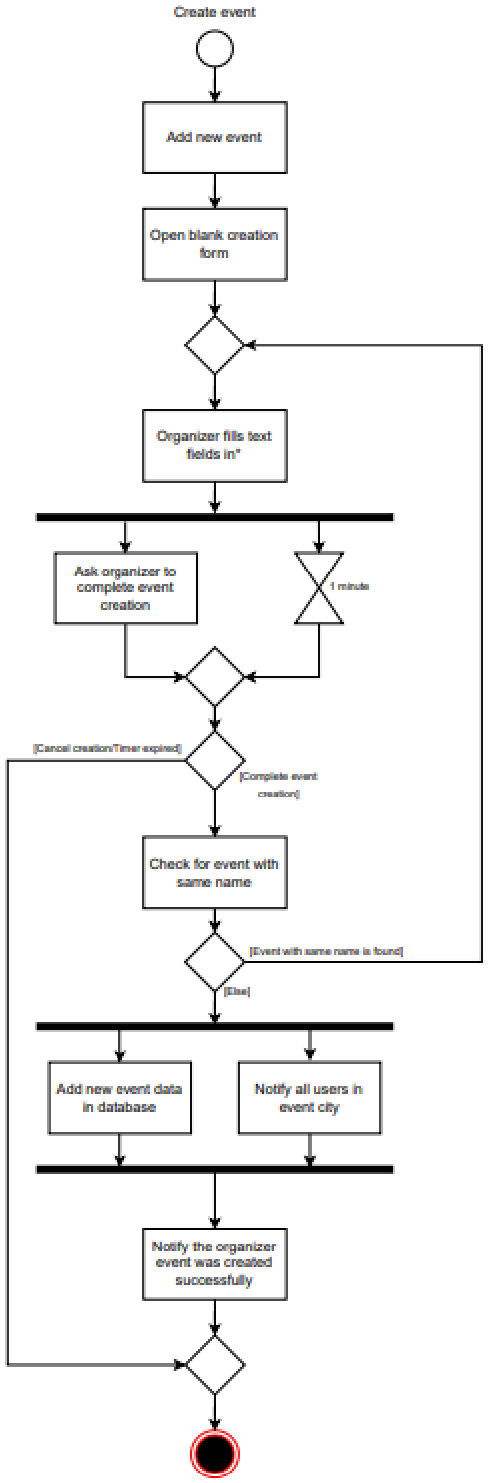
Matteo Della Giovampaola:

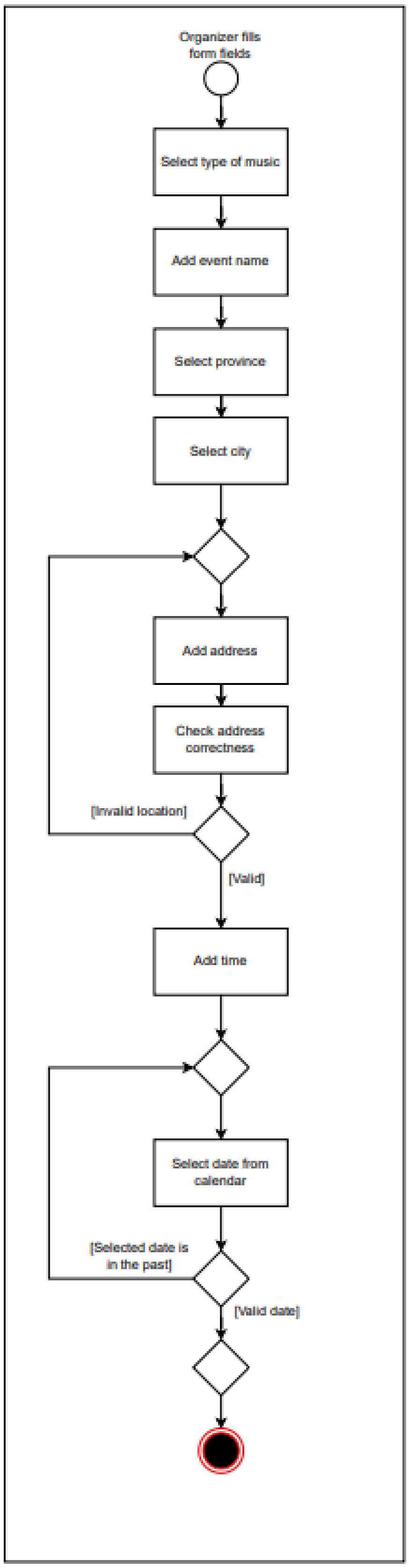
Use Case: *Plan event participation*



Emanuele Moscetta:

Use Case: *Create event*

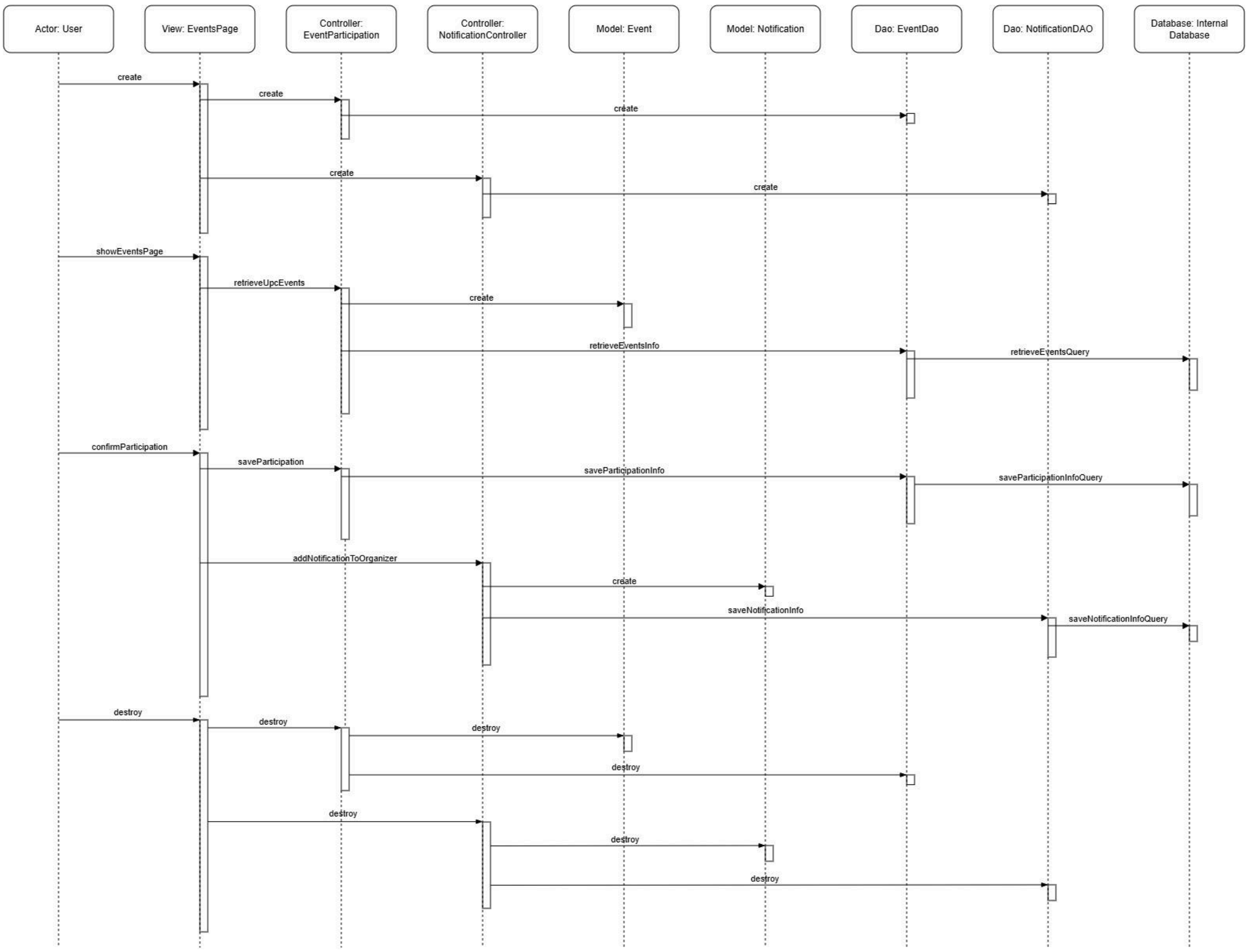




3.4 Sequence Diagram

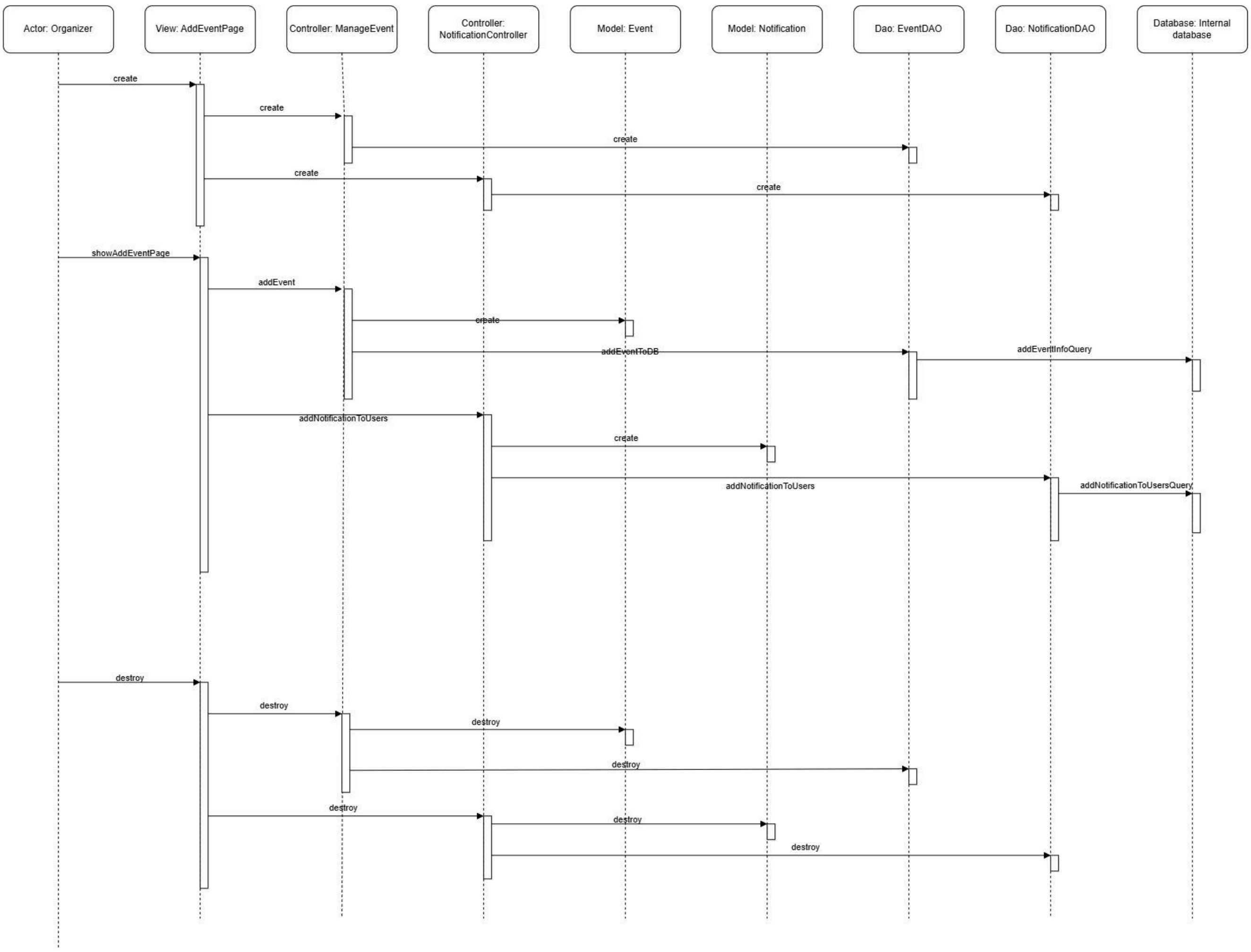
Matteo Della Giovampaola:

Use Case: *Plan event participation*



Emanuele Moscetta:

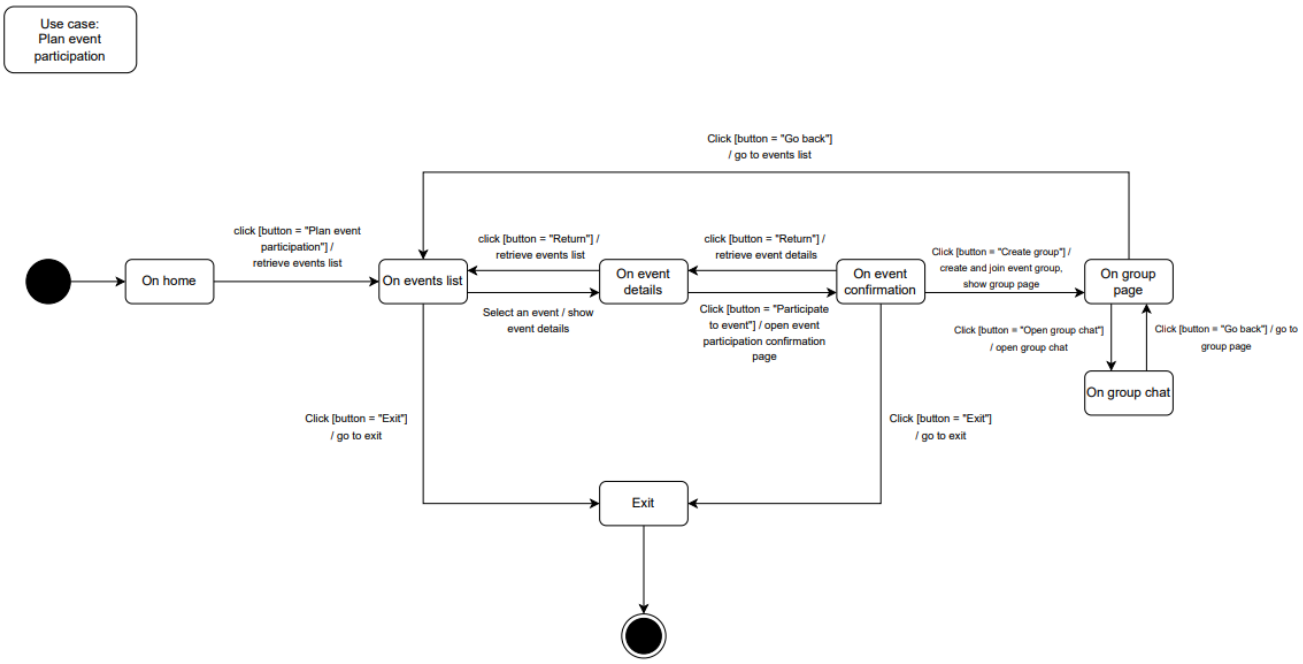
Use Case: *Create event*



3.5 State Diagram

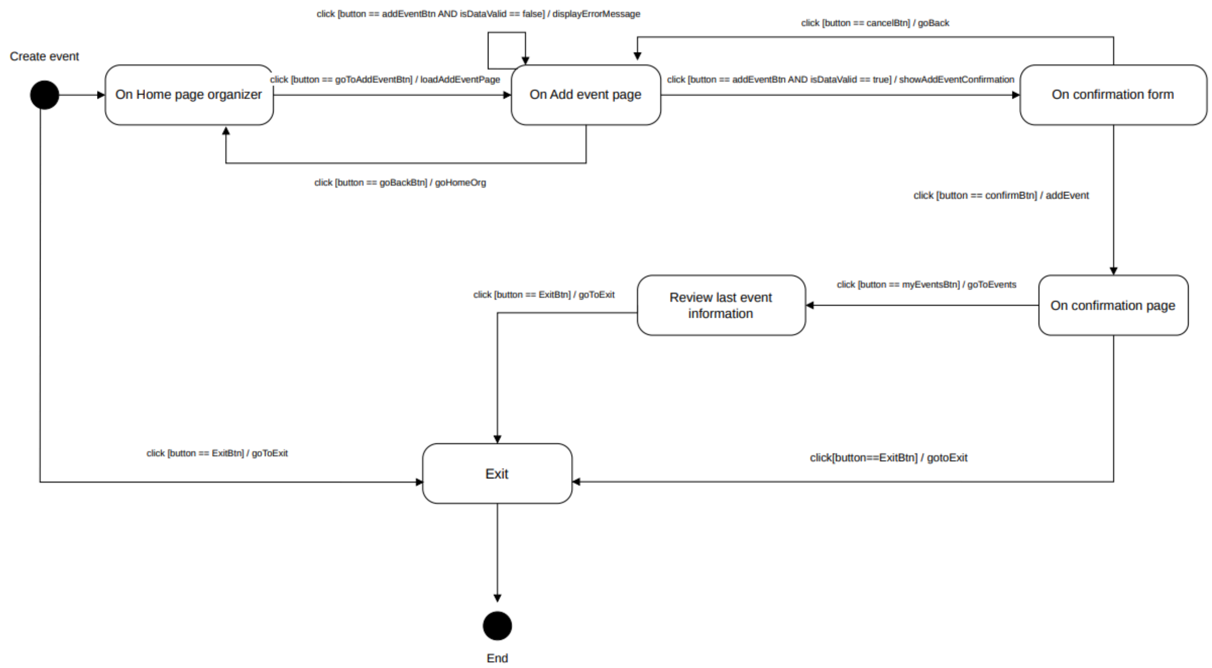
Matteo Della Giovampaola:

Use Case: *Plan event participation*



Emanuele Moscetta:

Use Case: *Create event*



3.6 Discrepancies

The following list outlines discrepancies identified between the final version of the application and its prototype.

Matteo Della Giovampaola:

1. **UseCaseSteps**: The system doesn't ask the user to participate. They system doesn't ask the user to create or join a group right after participating to the event.

2. **Activity Diagram**: Confirm event participation doesn't happen with confirmation link via email.

3. **State Diagram**: After event confirmation the user isn't immediately asked to create or join the event group.

Emanuele Moscetta:

4. **Use Case Steps**: Google Maps non utilized to check address correctness.

5. **Activity Diagram**: Address correctness isn't verified. The form doesn't auto close after 1 minute.

6. **State Diagram**: Confirmation page doesn't exist after confirmation form. After confirmation page there is no event recap, in the application we immediately go to your events page.

7. **Storyboards**: Filters and bookmarks missing in user settings.

4 Testing

To ensure the robustness and reliability of our application, we have developed six unit tests for our controllers using JUnit library. We divided the work among team members, with each person responsible for writing tests for three classes. This approach allowed us to effectively cover all key areas of the application. Here's an overview of how we implemented the test classes:

Division of Responsibilities

We divided the test classes between the two members of our group as follows:

Matteo Della Giovampaola:

TestManageEventController: used to test the creation of new event and the editing of a casual event previously created

TestNotificationController: used to verify the adding of a new notification

TestChatController: used to test the functionality of the chat controller, esuring the correct sending of a message

Emanuele Moscetta:

TestLoginController: used to verifiy the correctness of a new user registration and the change of city for a registered user

TestGroupController: used to test creation of a user group and joining of a group

TestEventParticipationController: used to test different functionalities related to the event participation, such as the participation of an event, the removal of the participation and the verification of a previous participation to the event

5 Exceptions

We have developed various custom exception classes in Java to handle error situations appropri- ately within our application.

The development of these custom exception classes was driven by the need for:

Specificity in Error Handling: Custom exceptions allow us to handle specific error sce- narios more precisely, providing meaningful error messages and responses to each situation.

Improved Code Readability: By defining our own exception classes, we make the code more readable and maintainable. It becomes clearer what types of errors are being handled and how they are addressed.

Enhanced Debugging: With specific exception types, it is easier to debug issues. The stack traces and error logs become more informative, leading in faster resolution of problems.

Below is a brief overview of the custom exception classes we have implemented:

1. **DuplicateEventParticipation**: this exception is thrown when a user have previously joined ad event.

2. **EventAlreadyAdded**: this exception is thrown when an organizer tries to create an event with the same name as another.

3. **EventAlreadyDeleted**: this exception handles the situation where an event is deleted by the organizer while a user, who previously joined it, is on Your Events page and tries to create a new group or join an existing group for the deleted event.

4. **GroupAlreadyCreated**: this exception is thrown when a user tries to create a new group for an event which already has a group. This situation happens when a user creates a group while another user has already completed the group creation process.

5. **InvalidGroupName**: this exception handles errors related to the creation of a group with an empty name tag.

6. **InvalidTokenValue**: this exception is used for error related to login with Google account operations. In particular, it is thrown when a user or an organizer put a wrong/invalid token given by Google to continue the login process.

7. **InvalidValueException**: this exception is thrown when a user inserts an invalid value in a form, such as an invalid date/time format.

8. **MinimumAgeException**: this exception is thrown during the registration process, when a user adds an age below 18 years old.

9. **TextTooLongException**: this exception is user to notificate users or organizers to reduce the amount of characters used in a textfield.

10. **UsernameAlreadyTaken**: this exception handles the situation when a user/organizer tries to register a new NightAgent account using a username which is already stored in our database.

Each project team member is responsible for handling five exceptions classes:

Matteo Della Giovampaola: DuplicateEventParticipation, GroupAlreadyCreated, InvalidGroupName, Invalid-TokenValue, UsernameAlreadyTaken

Emanuele Moscetta: EventAlreadyAdded, TextTooLongException, EventAlreadyDeleted, InvalidVal- ueException, MinimumAgeException

6 Databases

In the development of our system, we have decided to implement most of the Data Access Objects (DAO) to operate in JDBC (Java Database Connectivity) mode. The decision to use JDBC was driven by several considerations:

I. **Reliability and Performance:** JDBC is a well-established solution for interacting with relational databases, offering high performance and reliability.

II. **Scalability:** Using a relational database like MySQL enables the system to handle an increasing volume of data and requests without compromising performance.

III. **Maintainability:** JDBC provides a standardized method of database access, simplifying maintenance and support.

However, for the Notifications DAO, we have designed it to operate in dual modes: JDBC with MySQL and File System, storing data persistently using the .csv format. This dual implementation was considered to accommodate different use cases and operational flexibility:

! **JDBC Mode (MySQL):** This mode ensures that notification data is integrated with the rest of the system, taking advantage of MySQL's reliability, performance, and scalability.

! **File System Mode:** This mode offers a lightweight and easily deployable solution where a full database setup may not be necessary. Notifications are stored in a .csv file, providing a simple and human-readable format that can be used for quick data inspection and migration.

In a later development phase, we introduced a third persistence mode, in-memory, selectable at application startup through a dedicated Mode Selection screen ('Demo' vs 'Normal'). When Demo mode is selected, every DAO in the application (User, Location, Event, UserEvent, Group, Chat and Notification) uses a dedicated in-memory implementation instead of MySQL or the file system: data is kept in a shared runtime data store and is not persisted between executions. This mode was introduced to allow the application to be run and demonstrated without requiring a MySQL setup or existing data files, and does not replace the JDBC/File System modes described above, which remain the default and are still used for actual data persistence.

7 SonarCloud

By incorporating SonarCloud into our development workflow, we ensure that our application not only meets functional requirements but also adheres to the highest standards of code quality and security. You can check the results of SonarCloud analysis of this project using the following link:

https://sonarcloud.io/summary/overall?id=Matteo-dgsh_ProgettoISPW25-26&branch=mainSummary

VIDEO:

https://youtu.be/Y0aIUCC15os

GITHUB:

Matteo-dgsh/ProgettoISPW25-26